When Does Spatial–Spectral Fusion Help? A Functional-Identifiability Account with Computable Diagnostics and Multi-Order Interaction Decomposition

Panlin Li[a], Yuchang Li[a] and Hua Huang[a,b,*]

[a] *College of Mathematics and Physics, Xinjiang Agricultural University, Urumqi, 830052, China*
[b] *Center for Numerical Simulation and Data Analysis, Xinjiang Agricultural University, Urumqi, 830052, China*

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ABSTRACT

Spatial–spectral fusion is central to hyperspectral imaging (HSI), yet *when* and *why* it helps lack an a priori computable answer. Mainstream random-pixel splitting leaks spatially correlated pixels across train/test sets, systematically inflating accuracy. We model HSI cubes as function-valued spatial fields and decompose the optimal predictor into an additive part (marginal spectral + spatial) and an interaction residual C, whose energy γ bounds the fusion gain. Further decomposing γ into second-order (per-pixel spectral$\times$spatial) and higher-order components reveals the order-wise interaction structure; the second-order fraction ρ_2 indicates how close shallow additive-bilinear models approach the theoretical optimum. Cross-cohort generalization gap is characterized by a mapping drift Δ_{map} via Le Cam lower bounds. Leakage-safe spatial block cross-validation across four benchmarks yields: (1) interaction energy is second-order dominated (ρ_2=0.54$\sim$0.80); (2) a $\sim$270–480-parameter additive model reaches 73%$\sim$93% of the best deep model's macro F1 with 168$\sim$859$\times$ fewer parameters; (3) random splitting inflates OA by up to +38.6 pp relative to 6$\times$6 spatial blocking. $\hat{\gamma}$ and $\widehat{\Delta}$ are two pre-deployment computable diagnostics for fusion necessity and cross-scenario transferability.

1. Introduction

Hyperspectral imaging (HSI) simultaneously captures two-dimensional spatial images and hundreds of contiguous narrow spectral bands, producing tensor-valued data with coupled spatial and spectral structure [1, 2, 3]. Each pixel is a high-dimensional vector sampled from a continuous electromagnetic response; neighboring pixels are strongly correlated in space. HSI thus carries both *spectral* and *spatial* structural information. How to fuse these two modalities to improve accuracy, robustness, and interpretability is a central question in HSI analysis [4, 5]. Over the past decade, this has spurred increasingly complex architectures—from spatial–spectral composite kernels, Markov random fields, and morphological profiles, to 3D convolutions, spatial–spectral attention, and Transformers [6, 7, 8, 9, 10, 11]. These methods routinely report near-saturated accuracy under their respective data splits, fostering a default expectation that more complex fusion is always better [12, 13].

However, two foundational questions underpinning this belief remain unresolved.

Question 1 (Mechanism): Where does the gain of fusion over simple concatenation originate, and under what conditions does it exist? Can this gain be *decomposed* into an additive component and a non-additive interaction component, and can it be *computed a priori*? Existing work relies almost exclusively on end-to-end performance comparisons and lacks a certifiable, a priori computable characterization [19, 20]. Consequently, when a complex fusion model outperforms simple concatenation, one cannot determine whether the gain reflects genuine *non-additive spatial–spectral interaction* or merely stronger *additive* spatial-context modeling—two possibilities with radically different implications for model design and computational cost.

Question 2 (Evaluation validity): The mainstream protocol for HSI pixel classification uses random pixel splitting. This allows spatially adjacent—hence highly auto-correlated—pixels to appear in both training and test sets, causing information leakage that systematically inflates generalization estimates [29, 30, 31]. Under leakage-inflated evaluation, the perceived benefit of complex fusion may itself be an artifact of overestimation.

These two questions are intertwined: only under *leakage-free* evaluation with a *decomposable* framework can the true source and magnitude of fusion gain be disentangled. This paper addresses both questions through functional data analysis (FDA) [21, 22, 23]. We model each HSI cube as a *function-valued spatial field* $X(u, \lambda)$ over a spatial domain D and a continuous spectral domain Λ. We then formalize spatial–spectral fusion as an ANOVA-type decomposition of the optimal

*Corresponding author
huanghua@xjau.edu.cn (H. Huang)
ORCID(s): 0000-0003-0615-9402 (H. Huang)